SEMAHATTIN CAN DORUK



Software Developer | Flutter & Cross-Platform Mobile Development

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PROFESSIONAL SUMMARY

Software Developer specializing in Flutter and cross-platform mobile development. Built an industrial bioreactor control and monitoring platform through several stages - a C# / WPF desktop application backed by Supabase, a web version for remote control, and a full rebuild in Flutter - covering embedded communication, measurement data processing, and control logic. Also experienced with ESP32-based IoT development, ROS robotics integration, and backend work using Supabase, Firebase, PostgreSQL, and SQLite. At ESB, designed the product, UI/UX, and architecture of the earthquake-safety app ZATU and am building it end to end with Flutter, ASP.NET Core, and PostgreSQL/PostGIS. Published a Flutter application on Google Play. B.Sc. in Information Systems and Technologies, with an Erasmus+ exchange semester in Portugal.

TECHNICAL SKILLS

Languages: Dart, C#, Python, SQL, HTML/CSS

Mobile Development: Flutter, Dart, Cross-Platform Development, Android, iOS, State Management (Provider, Riverpod), go_router, Hive, Material Design, Material 3, Responsive UI, Google AdMob, In-App Purchase, Google Play Publishing

Desktop & Web: Flutter Desktop (Windows / Linux), C# / .NET, WPF, React, TypeScript, Next.js, WinForms, MVVM, Web-Based Control Interfaces, Asynchronous Programming, Multithreading

Backend & Databases: ASP.NET Core, Entity Framework Core, PostgreSQL, PostGIS, Supabase (Auth, RLS, Realtime), Firebase Firestore, SQLite, FastAPI, REST API Integration, JSON, Relational Data Modeling

Embedded Systems & IoT: ESP32, Raspberry Pi, NVIDIA Jetson Orin Nano, ROS (Robot Operating System), CMake, Serial Communication (UART), Bidirectional Communication Protocols, Measurement & Sensor Data Processing, Wi-Fi Access Point Mode, Linux Boot & Kiosk Configuration

Tools & Practices: Git, GitHub, GitHub Actions (CI), Docker, Automated Testing (Testcontainers, Widget Tests), k6 Load Testing, Linux, Software Architecture, Debugging, Code Review, Technical Documentation, Teamwork

Design & Data: UI/UX Design, Photoshop, Canva, Python Data Analysis, Machine Learning Fundamentals, Speech-to-Text Integration (faster-whisper)

PROFESSIONAL EXPERIENCE

Software Developer | ESB (Elektronik Sistem Blokları) | *Aug 2026 - Present*

Developing the ZATU mobile app end to end and delivering web projects ranging from a corporate CMS to marketing sites.

- **ZATU Mobile App (in development):** an earthquake and disaster-safety app built with the AFKOR disaster-coordination NGO; designed the product, UI/UX, and system architecture, and am developing every layer — Flutter (Riverpod, go_router), ASP.NET Core, EF Core, PostgreSQL/PostGIS, a background notification worker, and a React/TypeScript admin panel.
  - Built hold-to-alert emergency notifications, a mutual trusted-contact network, live location sharing during emergencies, photo damage reports, offline preparedness guides, and cross-device sync; added mandatory TOTP MFA, role-based access, and audit logging to the admin panel.
  - Verified the system with Testcontainers integration tests, widget and responsive-layout tests, GitHub Actions CI, k6 load tests, and backup-restore drills; made the whole environment runnable with a single Docker command.
- **Akkayalar A.Ş.:** Built an end-to-end corporate CMS and website with Next.js, TypeScript, Prisma, PostgreSQL, and Supabase — a full admin panel with news, documents, board members, org chart, committees, a media library, role-based user management, an audit log, and maintenance mode.
- **ZATU Website:** Built the app's public website with HTML, Tailwind CSS, and WordPress.
- **AFKOR:** Built the corporate website of a national disaster-coordination NGO with HTML, Tailwind CSS, Vanilla JavaScript, and Python.

Software Developer (Volunteer, Project-Based) | SANCAKTAR Mekatronik Sistemler | *Jul 2025 - Mar 2026*

Volunteer / project-based work on an industrial bioreactor control and monitoring platform.

- Built the bioreactor control and monitoring desktop application from scratch in C# and WPF, covering the UI, application logic, data processing, and hardware communication.
- Integrated the application with a Supabase backend for data storage, authentication, and centralized access to operational data.
- Developed a web version of the system so the bioreactor could be monitored and controlled remotely.
- Rebuilt the application from scratch in Flutter, moving it onto a single cross-platform codebase and reworking its architecture and UI.
- Implemented bidirectional serial (UART) communication between the application and the embedded control system.

- Processed incoming measurement data for control parameters including pH, redox, pO2, turbidity, stirrer, gas mix, and total flow, and ran the calculations used by the control logic.
- Sent the resulting control values back to the embedded system and implemented the control mechanisms that act on them.
- Implemented locking and control mechanisms within the system.
- Used MVVM and multithreaded background processing to keep the UI responsive during continuous data acquisition, along with structured error reporting.
- Set up the deployed Linux device with a custom boot screen, kiosk mode, and hardware / software locking that blocks external devices other than authorized flash drives.
- Worked in a 2-developer team, using Git and GitHub for parallel development.

Software Developer (Volunteer, Project-Based) | UGF Makine Sanayi ve Ticaret A.Ş. | *Jul 2025 - Jan 2026*
Volunteer / project-based work on IoT and robotics software.

- **Smart Sprinkler (IoT):** Designed and developed the complete product independently across ESP32 firmware, a Flutter mobile application, and a Supabase backend.
  - Set up Wi-Fi Access Point mode on the ESP32 so users could connect to the device from the mobile app and configure local network settings without on-site support.
  - Designed the Supabase / PostgreSQL data model with Row-Level Security, authentication, and realtime data streaming, and integrated the device, backend, and mobile app.
- **Weeding Robot:** Handled the software integration and designed the operator control interface for an autonomous agricultural robot.

Software Developer Intern | Nefel Hassas Tarım | *Jun 2025 - Aug 2025*

- Developed and integrated control software for autonomous agricultural robots and a solar-panel cleaning robot.
- Set up and configured ROS (Robot Operating System) environments on NVIDIA Jetson Orin Nano and Raspberry Pi, and integrated them with the robot hardware.
- Managed builds and configuration with CMake, and integrated serial communication protocols for robot motion control.
- Designed sensor-based headland (U-turn) navigation logic for autonomous field operation.
- Documented installation and configuration steps for field deployment.

Python Instructor (Part-Time) | Kodland | *Apr 2024 - Jul 2024*

- Taught Python, algorithmic thinking, and software fundamentals to 5 students across 2 premium groups, adapting lessons and exercises to each student's level.

SELECTED PERSONAL PROJECTS

Habitly Daily - *Flutter, Dart, Riverpod, Hive, AdMob, In-App Purchase*
Habit-tracking app with reminders, focus sessions, and offline local storage. Integrated Google AdMob and in-app purchases. Published on Google Play.
play.google.com/store/apps/details?id=com.drkapps.habitlydaily

CrestBalance - *Flutter, Supabase, PostgreSQL, Riverpod*
Multi-currency budget and grant tracking app for Erasmus students, with authentication and realtime sync built on Supabase Row-Level Security.

Voice2 Note AI - *Flutter, FastAPI, SQLite, faster-whisper*
Note-taking app that transcribes audio recordings and generates summaries. The Flutter client runs against a self-hosted FastAPI service.

AgroMora - *Flutter, Supabase, PostgreSQL*
Mobile e-commerce app with role-based access control (RBAC) across separate user roles.

BağışYap - *Flutter, Firebase Firestore, Provider, Material 3*
Donation app built with Flutter and Firebase Firestore, using Provider for state management and Material 3 for the interface.

EDUCATION

B.Sc. Information Systems and Technologies | Muş Alparslan University, Türkiye | 2022 - 2026 | GPA 3.10 / 4.00

Erasmus+ Exchange Programme | Universidade da Beira Interior, Portugal | Spring 2026

Web Design and Programming | Yahya Günsür Vocational and Technical Anatolian High School | 2016 - 2020

ADDITIONAL INFORMATION

Languages: Turkish (Native), English (C1)

Community: Team Member, TEDx Muş Alparslan University - event operations and inter-community coordination.

Portfolio & Code: github.com/CanDorukk | candorukk.github.io